Subsidising the Metro per passenger or per kilometre? A calibrated translog cost function for the Barcelona Metro (FMB-TMB), 2023–2025

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June 9, 2026

Abstract

This paper calibrates the cost structure of the Barcelona Metro (Ferrocarril Metropolità de Barcelona, FMB) for 2023–2025 using a multiproduct translog cost function on a line-year panel of 21 observations (7 operating blocks $\times$ 3 fiscal years). Two items in the FMB accounts—the Ifercat L9 concession fee and train leasing—jointly represent about 29 per cent of total declared cost and have the character of quasi-fixed contractual transfers rather than technological inputs. We therefore model total cost as the additive decomposition $C_{\text{total}}(Y, W, \text{inst.}) = C_{\text{tec}}(Y, W) + F_{\text{inst}}$, where C_{tec} is the translog cost function and F_{inst} is the institutional component. Forcing F_{inst} inside the translog breaks Shephard concavity in 48 per cent of observations, a direct empirical confirmation of the conceptual argument. Under the calibrated technological model, returns to density are substantial (RTD $\approx$ 2.18), under the calibrated parameters the marginal cost is about 0.42 €/pax against an average cost of 0.91 €/pax (MC/AC $\approx$ 0.46), and the Boiteux optimal subsidy is approximately 54 per cent of technological average cost. F_{inst} contributes roughly 0.38 €/pax system-wide and reflects institutional burden rather than operational inefficiency: it represents around 70 per cent of total cost in L9/L10 but only 8 per cent in L1–L5. Accounting separation aligned with the spirit of Directive 2012/34/EU is not applied in FMB consolidated accounts, biasing any efficiency or social cost–benefit assessment.

Keywords: Translog cost function; urban rail; returns to density; infrastructure access charge; optimal subsidy; Barcelona.
JEL classification: L92, L43, R41, H42.

1 Introduction

The economics of urban metro systems combines three features that the literature regards as central for regulatory design: network structure, sunk capital investment, and substantial returns to density. These features justify operating subsidies, tariff regulation, and dedicated concession contracts (Joskow, 2008; Gagnepain and Ivaldi, 2002). Ferrocarril Metropolità de Barcelona, S.A. (FMB), the operating subsidiary of the Barcelona Metro within the Transports Metropolitans de Barcelona (TMB) group, exhibits an additional institutional feature of regulatory relevance: lines L9 and L10—among the longest driverless automated metro lines in Europe—are operated under an availability-based concession fee paid to the public entity Infraestructures Ferroviàries de Catalunya (Ifercat), a body of the Generalitat de Catalunya that owns the underlying infrastructure. In 2023–2024 this fee exceeded 135 M€ per year, equivalent to 22–23 per cent of total recurrent cost, and it is designed to recover approximately 8 billion € of historical civil-works investment over the asset life. L9/L10 carries roughly 6 per cent of system passengers but generates the single largest institutional cost item in the group accounts. A second contractual item—train leasing (45–47 M€

per year, around 6–7 per cent of total cost)—reflects operationalised capital cost via long-term financial-lease agreements, mainly for the renewal of the 6000/7000 fleet on lines L1, L3 and L5.

Despite the regulatory centrality of Barcelona within Spain, the empirical literature has not quantified the elasticities of its cost function with a methodology that can inform tariff or concession reform: Matas and Raymond (2009) examines Spanish urban transport at the national level without isolating the rail concession fee, and Daraio et al. (2016) reviews international metro frontiers without a Barcelona case. The contribution of this paper is not to replicate the existence of density economies, but to isolate the technological component of the cost function from the institutional component and to demonstrate microeconomically that the two must be analysed separately. We calibrate a translog cost function $C_{\text{tec}}(Y, W)$ over the line-year panel under expenditure minimisation, treat the contractual items (concession fee, leasing) as an additive non-functional component F_{inst}, and show that forcing F_{inst} inside the translog produces concavity failure in 48 per cent of observations and a 3.8× degradation in fit—direct evidence that this is not a technological cost.

European rail-sector regulation has, since the early 1990s, progressively required vertical separation between infrastructure managers and operators, culminating in Directive 2012/34/EU (Directive 2012/34/EU, 2012), which establishes the single European railway area and mandates accounting separation between the two functions. Urban metro systems are not formally within the scope of this directive, but its underlying economic logic—that mixing infrastructure-related charges with operating costs distorts efficiency signals and biases regulatory comparisons—applies a fortiori to systems such as TMB where the infrastructure-access charge represents nearly a quarter of total declared cost. This paper provides the microeconomic foundation for that logic and quantifies what it implies for the Barcelona Metro.

The paper is organised as follows. Section 2 sets out the theoretical framework and the connection with network regulation. Section 3 describes the data and the construction of the panel. Section 4 presents the calibration strategy, including the treatment of non-technological components (Section 4.5) and the production technology and substitution margins (Section 4.6). Section 5 reports the results under the four hypotheses, with the decomposition evidence in Section 5.4. Section 6 consolidates the robustness analyses. Section 7 discusses regulatory and policy implications. Section 8 acknowledges limitations; Section 9 concludes.

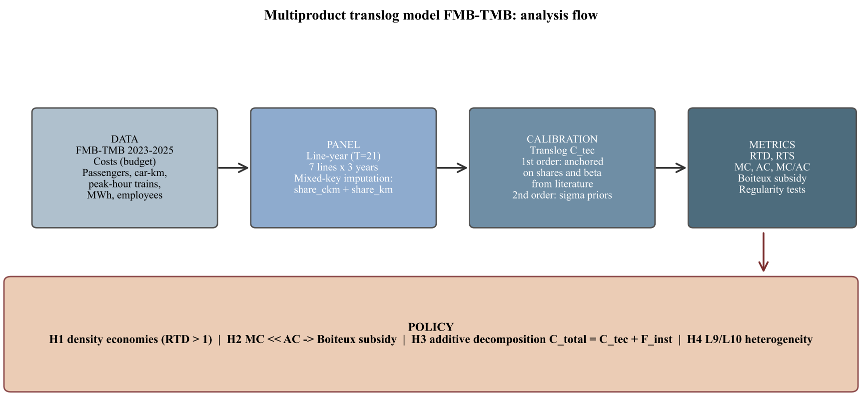


Figure 1: Methodological flow: data → line-year panel → calibration → metrics → policy.

2 Theoretical framework

2.1 Multiproduct translog cost function

The translog cost function (Christensen et al., 1973) is a second-order approximation to an arbitrary cost function. With two outputs—passengers Y_1 and service car-km Y_2—and three inputs—labour

L, energy E and an aggregate of other inputs O—the specification is

$$\ln C = \alpha_0 + \sum_{i\in\{L,E,O\}} \alpha_i \ln W_i + \sum_{k=1,2} \beta_k \ln Y_k + \tfrac{1}{2}\sum_{i,j}\gamma_{ij}\ln W_i \ln W_j + \tfrac{1}{2}\sum_{k,l}\delta_{kl}\ln Y_k \ln Y_l + \sum_{i,k}\rho_{ik}\ln W_i \ln Y_k + \lambda t + \varepsilon. \quad (1)$$

Linear homogeneity in prices imposes $\sum_i \alpha_i = 1$, $\sum_j \gamma_{ij} = 0$ and $\sum_i \rho_{ik} = 0$. The *symmetry* restrictions $\gamma_{ij} = \gamma_{ji}$ and $\delta_{kl} = \delta_{lk}$ are assumed. Normalising by the price of the *other-inputs* aggregate W_O (the numeraire) and centring all variables on the sample geometric mean, the cost equation is rewritten in terms of the relative prices W_L/W_O and W_E/W_O only.

2.2 Returns to density, scale and scope

Following Caves et al. (1984), returns to density are defined as the reciprocal of the cost–passenger elasticity (network held fixed):

$$\mathrm{RTD} \;=\; \frac{1}{\partial \ln C/\partial \ln Y_1} \;=\; \frac{1}{\varepsilon_{Y_1}}, \quad (2)$$

and returns to scale as $\mathrm{RTS} = 1/(\varepsilon_{Y_1} + \varepsilon_{Y_2})$. For a translog function the elasticities *vary by observation* because of the interaction terms δ_{kl} and ρ_{ik}. A value $\mathrm{RTD} > 1$ indicates increasing returns to network *utilisation*: adding passengers to a given car-km supply reduces average cost.

2.3 Cost shares and Shephard's lemma

By the lemma of Shephard (1953) and the cost–production duality of Diewert (1971), input cost shares satisfy $s_i = \partial \ln C/\partial \ln W_i$ and, in a translog,

$$s_i \;=\; \alpha_i + \sum_j \gamma_{ij}\ln W_j + \sum_k \rho_{ik}\ln Y_k. \quad (3)$$

The Allen–Uzawa elasticities of substitution (Uzawa, 1962) are then $\sigma_{ij} = 1 + \gamma_{ij}/(s_i s_j)$ for $i \neq j$.

2.4 Connection with network regulation

When marginal cost is well below average cost—a typical feature of network industries—marginal-cost pricing generates structural deficits. The rule of Boiteux (1956) establishes that the optimal tariff depends on the elasticity of demand, but the *level* of the public subsidy required to close the gap is $1 - \mathrm{MC}/\mathrm{AC}$. This logic is part of the body of public-finance theory of urban transport (Small and Verhoef, 2007; Pigou, 1920; Allais, 1947) and is the foundation of the metropolitan operating subsidy in Barcelona.

3 Data and panel construction

3.1 Sources

Financial figures come from the quarterly tracking of the FMB budget published by the TMB transparency portal for 2023 (Q4), 2024 (Q4) and 2025 (Q3, partial), cross-checked with the audited FMB 2023 accounts and Note 18 of the audited 2024 TMB Group report (which isolates the FMB subsidiary). Operating indicators—passengers per line, car-km, place-km, employees, peak-hour trains and network length—come from the TMB *Basic data* fact sheets. Physical electricity consumption is taken from the 2024 TMB Sustainability Report (255,413 MWh in 2023; 239,402 MWh in 2024).[1]

[1] Operating labour in the audited accounts (240.482 M€ in 2023) differs by 0.5 per cent from operating labour in the budget (239.328 M€). We use the budget series for temporal consistency with 2024 and 2025.

3.2 Line-year panel

The granularity of the published financial statements is the consolidated subsidiary (FMB), not the individual line. With three years of aggregate data the variability matrix of the translog function is structurally ill-conditioned (see Section 6). We therefore construct a *line-year panel* of 21 observations by mixed-key imputation of the FMB cost categories to 7 operating blocks: L1, L2, L3, L4, L5, L9/L10 North and L9/L10 South. L11 and the Funicular are excluded on grounds of technological heterogeneity.

3.3 Theoretical justification of the imputation key

The choice of imputation key is the central methodological decision. Small and Verhoef (2007) and Nash and Preston (1995) argue that operating costs of urban rail—driving, supervision, light maintenance, traction energy—scale with *service supply* (car-km), not with realised demand (passengers). Imputing every cost item proportionally to passengers would mechanically yield a cost–passenger elasticity equal to unity and RTD = 1. We therefore adopt:

- Operating labour, energy, supplies, external services and train leasing are imputed by the car-km share, $\text{share\_ckm}_{l,t} = \text{peak\_trains}_{l,t} \times \text{km\_line}_l$, normalised at the system level.
- Net depreciation is imputed by the share of network length (*share_km*), reflecting installed physical capital.
- The Ifercat L9 concession fee is allocated entirely to L9/L10, split between the North and South blocks in proportion to their passenger share.

3.4 Treatment of 2025 (partial)

Published FMB accounts for 2025 cover January–September. We annualise all items by multiplying by 12/9. For energy, the Q3 €/kWh price is not a reliable representation of the annual closing figure; we impute the 2024 MWh and price (0.0815 €/kWh; 239,402 MWh). The workforce is not annualised (it is an end-of-period stock).

3.5 Variable definitions

Table 1: Operating definitions.

Variable	Definition
C_{tec}	Technological cost: labour + energy + supplies + external services + net depreciation.
F_{inst}	Institutional component: train leasing + Ifercat L9 concession fee.
C_{total}	Total declared cost: $C_{\text{tec}} + F_{\text{inst}}$.
Y_1	Passengers (entries to the network, by block).
Y_2	Imputed useful car-km.
W_L	Annual FMB operating labour cost / Metro headcount (€/employee).
W_E	Annual FMB energy cost / electricity consumption (€/kWh).
W_O	Numeraire (price of the other-inputs aggregate $\equiv 1$).
t	Centred time trend (year $-$ 2024).

3.6 Accounting coverage

The sum of imputed costs across the 7 included lines covers 99.8 per cent of total declared cost (C_{total}) and 99.7 per cent of technological cost (C_{tec}). The remaining 0.2–0.3 per cent corresponds to L11 and the Funicular, which are excluded from the panel but remain in the aggregate figures. The accounting reconciliation is exact by construction.

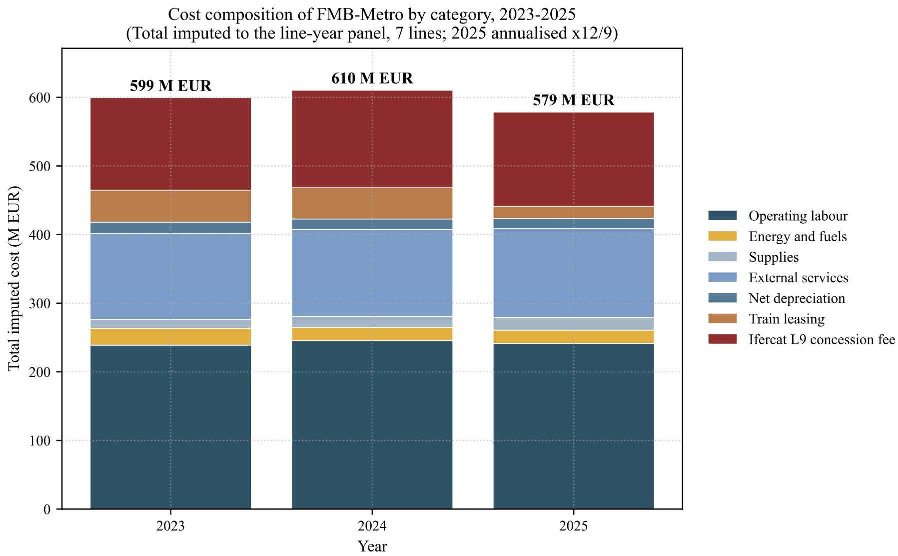


Figure 2: Cost composition of FMB-Metro by category and year.

3.7 Descriptive statistics

Table 2 summarises the statistics of the line-year panel ($N = 21$).

Table 2: Descriptive statistics of the line-year panel, $N = 21$.

Variable	Mean	Std. dev.	Min.	Max.
C_{total} (M€)	84.55	24.15	50.28	124.82
C_{tec} (M€)	60.24	26.93	26.59	109.05
Operating labour (M€)	33.79	16.25	14.94	62.81
Energy (M€)	2.93	1.41	1.29	5.45
Supplies (M€)	2.10	1.03	0.88	4.11
External services (M€)	16.24	7.85	7.29	32.30
Net depreciation (M€)	2.07	0.74	1.13	3.06
Train leasing (M€)	4.80	2.29	1.15	11.65
Ifercat concession fee (M€)	18.57	35.93	0.00	84.72
Passengers (M)	62.55	41.42	11.04	126.39
Imputed car-km (M)	13.97	7.13	6.01	26.21
W_L (€/employee)	60,092	1,166	59,108	61,298
W_E (€/kWh)	0.0864	0.0073	0.0815	0.0963

4 Calibration strategy

4.1 Calibration vs. econometric estimation

With $N = 21$, free estimation of the 17 parameters of the biproduct, three-input translog is infeasible: the effective degrees of freedom are low and input prices vary only at the system-year level. We therefore adopt a strategy of *anchored calibration*, in the methodological spirit of network regulation (Joskow, 2008; Angrist and Pischke, 2014) and following the recommendations of Coelli et al. (2005) for efficiency analysis with limited data. First-order parameters are anchored on observed moments; second-order parameters are anchored on external priors from the urban rail literature.

4.2 First-order anchoring

Mean input shares on total cost are weighted by cost across the 21 observations (equivalent to the aggregate imputed sum). For the technological cost function C_{tec}: $\alpha_L = 0.574$, $\alpha_E = 0.050$,

$\alpha_O = 0.376$. First-order cost–output elasticities are anchored on Savage (1997) and Graham (2008): $\beta_1 = 0.45$ and $\beta_2 = 0.25$.

4.3 Second-order priors

Allen–Uzawa elasticities of substitution are taken from Wheat and Smith (2015): $\sigma_{LE} = 0.40$, $\sigma_{LO} = 0.60$, $\sigma_{EO} = 0.50$. These values are narrow but non-zero, consistent with substitution operating through limited technological margins (see Section 4.6). The γ_{ij} are derived from $\gamma_{ij} = s_i\, s_j(\sigma_{ij} - 1)$ and the γ_{ii} by homogeneity. Output–output interactions are $\delta_{11} = \delta_{22} = 0.05$, $\delta_{12} = -0.05$; price–output interactions are $\rho_{LY_1} = -0.04$, $\rho_{LY_2} = +0.02$, $\rho_{EY_1} = +0.01$, $\rho_{EY_2} = +0.01$, with ρ_{OY_k} pinned by homogeneity. The technical trend is $\lambda = -0.005$. The full list of calibrated values is reported in Table 3.

4.4 Regularity tests

A well-behaved cost function must satisfy monotonicity in prices and outputs and concavity in prices. The calibrated technological function C_{tec} satisfies *monotonicity in prices* and *monotonicity in outputs* in all 21 observations and *approximate concavity* (maximum eigenvalue of the substitution matrix $M = H + ss' - \text{diag}(s)$ below 10^{-3}), with a maximum eigenvalue of $+9.3 \times 10^{-5}$ (numerical noise).

4.5 Treatment of non-technological components

FMB's accounts include two items—the Ifercat L9 concession fee and train leasing—that do not meet the assumptions underpinning the translog cost function. A cost function arising from expenditure minimisation must satisfy Shephard's (1953) properties (monotonicity, concavity in prices, linear homogeneity, symmetry), which derive from duality with a quasi-concave production function and from the operator's optimisation. The Ifercat fee violates this framework on three simultaneous grounds: (i) it does not depend on outputs, being a multi-year contractual payment; (ii) it does not depend on input prices, being independent of wages or electricity; (iii) it is not an outcome of optimisation by FMB, which can only accept the fee.

Train leasing admits a softer analogous argument. Rolling stock is a productive input, but its effective price to FMB is set by long-term financial-lease contracts with a fixed multi-year payment schedule that does not respond to year-by-year decisions and cannot be optimised in the short run. This makes the leasing payment closer to a contractually fixed transfer than to a variable input price; including it in C_{tec} would impute optimisation to a payment FMB cannot optimise. We therefore place it in F_{inst}, a conservative treatment (Section 8): under an alternative capital-cost modelling, average cost would shift marginally but marginal cost would remain that of C_{tec}.

For these reasons we model total cost as an additive decomposition:

$$C_{\text{total}}(Y, W, \text{institutions}) \;=\; C_{\text{tec}}(Y, W) + F_{\text{inst}}, \tag{4}$$

where $C_{\text{tec}}(Y, W)$ is the translog function (1), satisfying the standard properties, and F_{inst} = concession fee + train leasing is an additive non-functional component, accounted for but not modelled as technology. This formulation has precedents in network regulation (Joskow, 2008) and transport specifically: Gagnepain and Ivaldi (2002) apply an analogous separation in urban bus concessions, Mizutani and Uranishi (2013) distinguish infrastructure from operating costs in Japanese railways, and Wheat and Smith (2015) warn that pooling heterogeneous cost components into a single translog distorts the inferred returns to scale—a logic we extend to institutional heterogeneity.

As a complementary empirical diagnostic, in Section 5.4 we show that if F_{inst} were forced inside the translog specification, concavity would fail in 10 of 21 observations (48 per cent of the sample), with a maximum eigenvalue of the Allen–Uzawa substitution matrix of $+0.0081$. This is the expected microeconomic outcome and confirms that the additive separation is not optional but theoretically required.

4.6 Production technology and substitution margins

The cost function in (1) is calibrated at the line-year level. At the train-trip level the technology is essentially Leontief: a conventional train requires exactly one driver, a fixed electrical consumption set by the rolling stock, and a fixed maintenance intensity per car-km—five drivers cannot substitute for one, and a kilowatt-hour cannot replace a driver (Small and Verhoef, 2007). Substitution emerges only at the line-year level, where the operator aggregates many train-trips and chooses between alternative technologies, vintages and policies. Three margins are economically relevant and empirically active in FMB during 2023–2025:

(a) **Automation.** L9 and L10 operate driverless under CBTC signalling, whereas L1–L5 operate with one driver per train. This is a direct substitution of capital for labour at constant output. The progressive expansion of L9/L10 service hours over the period reflects this margin in operation.

(b) **Fleet renewal.** The introduction of series 6000/7000 rolling stock on L1, L3 and L5 reduces electrical consumption per car-km while delivering the same service. This is energy–capital substitution at constant output.

(c) **Make-or-buy in maintenance.** The proportion of maintenance performed in-house versus outsourced to manufacturers (Alstom, CAF, Siemens) can be adjusted, substituting in-house labour for external services without altering the output produced.

These margins generate the substitutability the translog captures. They are narrow—the calibrated Allen–Uzawa elasticities of approximately 0.4–0.6 reflect precisely this narrowness—but non-zero, and sufficient for the substitution matrix to exhibit the curvature required by Shephard duality. A fourth potential margin—adjusting service frequency—is not a clean substitution channel, since it shifts unobserved service quality (occupancy, waiting time, congestion) rather than substituting factors at constant output. We adopt the standard convention of treating Y_2 as a proxy that bundles frequency with car-km (Farsi et al., 2006; Daraio et al., 2016), returning to its implications in Section 8.

Crucially, the Ifercat L9 concession fee is impermeable to all three genuine margins: no automation, fleet-renewal or make-or-buy decision alters a fee fixed by multi-year contract with the Generalitat. The concavity failure documented in Section 5.4 is therefore not a generic departure from cost minimisation but the direct consequence of forcing into a cost function a component impervious to the only substitution channels operating in the technology. The additive decomposition (4) formalises this fact: substitutable components belong inside the cost function and generate concavity; non-substitutable contractual transfers belong outside it.

5 Results

5.1 Calibrated parameters of C_{tec}

Table 3 reports the calibrated parameters of the technological cost function C_{tec}.

Table 3: Calibrated parameters of C_{tec}. The first block contains first-order parameters (anchor shares and output elasticities); the second block contains second-order parameters pinned by homogeneity and the Allen–Uzawa priors.

Parameter	C_{tec}	Source of the prior
α_L	0.574	Data (labour cost share)
α_E	0.050	Data (energy cost share)
α_O	0.376	$1 - \alpha_L - \alpha_E$
β_1	0.45	Savage 1997, Graham 2008
β_2	0.25	Savage 1997
λ	-0.005	Observed technical trend
γ_{LL}	$+0.1036$	Homogeneity
γ_{EE}	$+0.0267$	Homogeneity
γ_{OO}	$+0.0957$	Homogeneity
γ_{LE}	-0.0173	$\sigma_{LE} = 0.40$ (W&S 2015)
γ_{LO}	-0.0863	$\sigma_{LO} = 0.60$
γ_{EO}	-0.0094	$\sigma_{EO} = 0.50$
δ_{11}	$+0.05$	Modest curvature
δ_{22}	$+0.05$	Modest curvature
δ_{12}	-0.05	Weakly complementary outputs
ρ_{LY_1}	-0.04	Literature prior
ρ_{LY_2}	$+0.02$	Literature prior
ρ_{EY_1}	$+0.01$	Adjusted for frequency rigidity
ρ_{EY_2}	$+0.01$	Literature prior

5.2 H1 — Density economies

Under C_{tec}, the passenger-weighted RTD of the system is **2.18**, comfortably above unity: doubling passengers on the existing network would reduce technological average cost by approximately 54 per cent. Heterogeneity is marked: lines L1–L5 exhibit RTD in the range 2.13–2.19, whereas L9/L10 (North and South) reach 2.36–2.43. The reason is that the automated lines operate with a rigid frequency and low occupancy, so each additional passenger absorbs idle capacity.

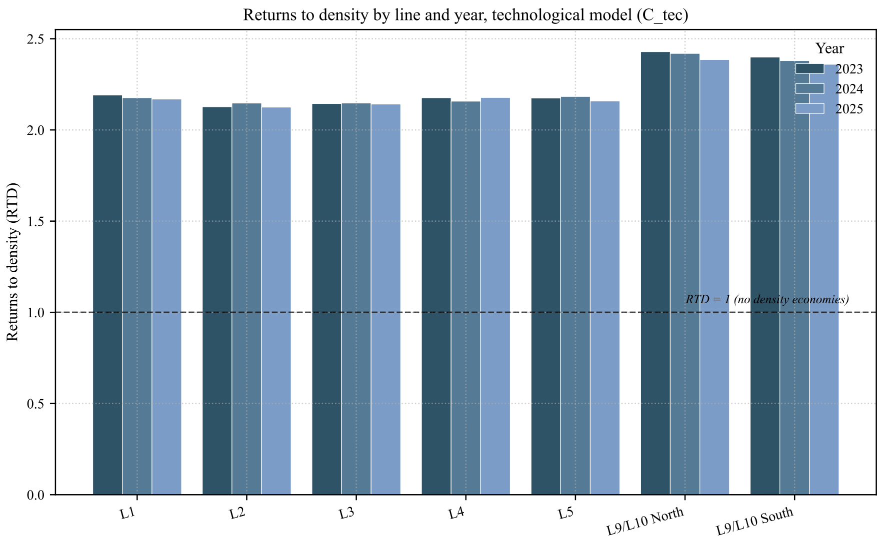
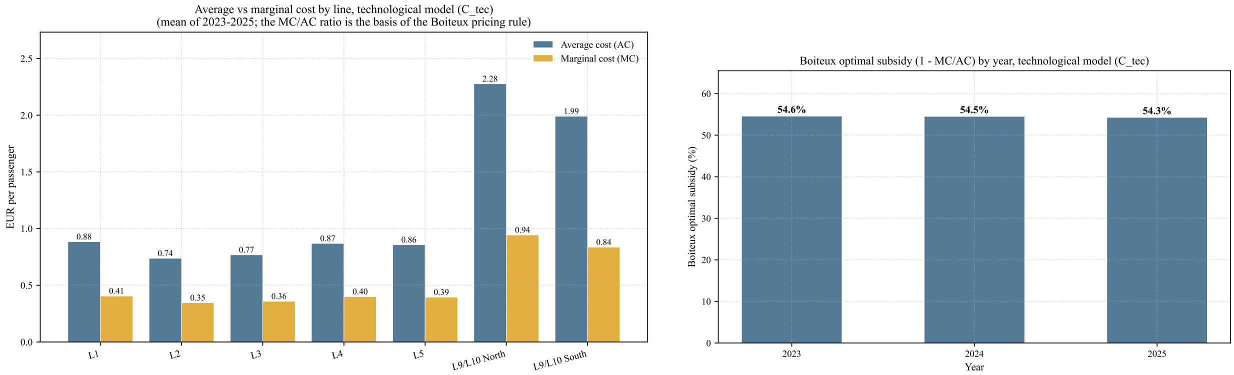


Figure 3: Returns to density by line-year, technological cost function C_{tec}.

5.3 H2 — Marginal vs. average cost

The passenger-weighted technological average cost of the system is 0.915 €/pax; the corresponding marginal cost is 0.417 €/pax. The MC/AC ratio is **0.46**, implying a Boiteux optimal subsidy of **54.5** per cent of technological average cost. Figure 4 reports the line averages (left) and the temporal evolution (right); the Boiteux optimal subsidy is stable across the three years.



(a) AC vs. MC by line, 2023–2025 mean. **(b)** Boiteux optimal subsidy by year.

Figure 4: Average vs. marginal cost and the implied Boiteux optimal subsidy, technological cost function C_{tec}.

5.4 H3 — Technology vs. institutional decomposition

(a) Magnitude and composition of F_{inst}

F_{inst} accounts for approximately 29 per cent of total declared cost. Under the calibrated parameters this corresponds to about 0.38 € per passenger system-wide, on top of the technological average cost of 0.91 € per passenger. Table 4 summarises the closing decomposition.

Table 4: H3 closing decomposition: $C_{\text{total}} = C_{\text{tec}} + F_{\text{inst}}$. Aggregates use passenger-weighted means over the panel.

Metric	C_{tec} (technological)	F_{inst} (institutional)	C_{total} (full)
Mean AC (€/pax)	0.91	0.38	1.29
Mean MC (€/pax)	0.42	0.00	0.42
Returns to density (RTD)	2.18	N/A (fixed)	N/A
Cost share	71%	29%	100%
Satisfies Shephard's properties	Yes	Not applicable (non-technological)	No (concavity fails in 10/21 obs.)

F_{inst} is overwhelmingly concentrated in L9/L10: those two lines absorb 82 per cent of the institutional component, and for them F_{inst} represents 70 per cent of total declared cost. By contrast, F_{inst} is only 8 per cent of total cost in lines L1–L5. This concentration is a direct consequence of how the concession fee is allocated by construction (100 per cent to L9/L10 by the Ifercat contract); the train-leasing component is more evenly spread.

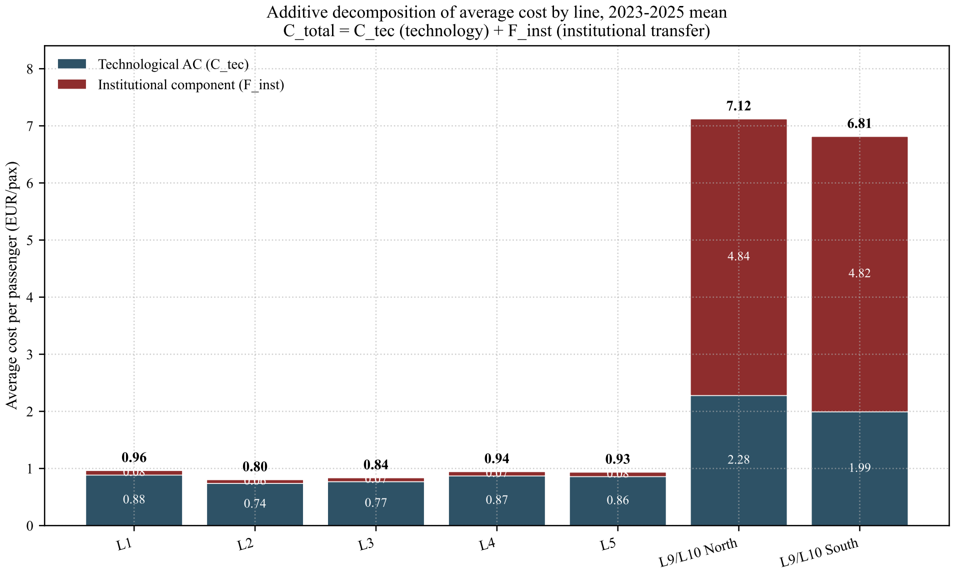


Figure 5: Additive decomposition of average cost by line, 2023–2025 mean. The dark band is technological average cost (C_{tec}); the red band is the institutional component (F_{inst}). The visible total is C_{total}.

(b) Convergent technological cost across lines

Across the seven blocks the technological average cost is concentrated in a narrow range, approximately 0.7–1.0 € per passenger. Total declared average cost in L9/L10 reaches 6–8 € per passenger. The gap between the two figures is entirely accounted for by F_{inst}, not by operational inefficiency. Once the institutional component is removed, the lines look technologically similar; the divergence reappears only when the concession fee is included.

(c) Concavity diagnostic

Forcing F_{inst} inside the translog specification breaks Shephard concavity. Figure 6 reports the maximum eigenvalue of the Allen–Uzawa substitution matrix for each of the 21 line-year observations: 10 observations (48 per cent) yield a positive eigenvalue, with a maximum of +0.0081, concentrated in L9/L10 where the fee weighs most heavily on the cost share.

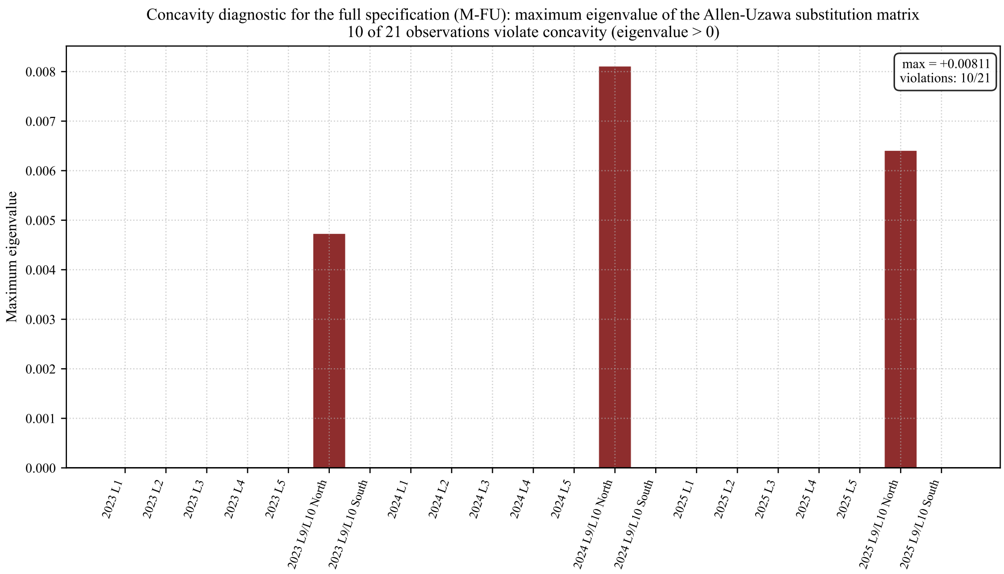


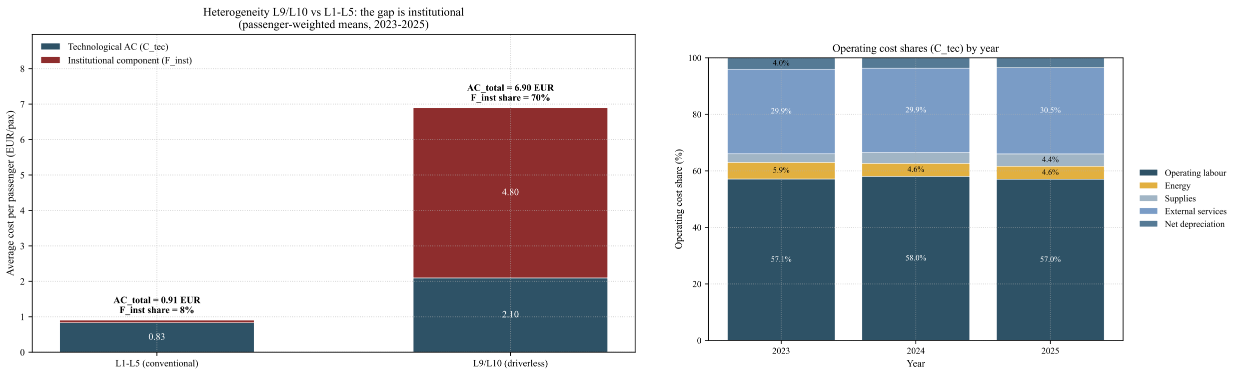
Figure 6: Concavity diagnostic for the full specification: maximum eigenvalue of the Allen–Uzawa substitution matrix, by line-year. Observations with eigenvalue > 0 violate Shephard concavity.

This failure is the expected microeconomic result. As argued in Section 4.6, the concession fee is impermeable to the three substitution margins (automation, fleet renewal, maintenance make-or-buy) that generate the concavity property in the technological cost function. A function that includes F_{inst} is asked to represent as technology something that is not technology, and the substitution matrix loses the required curvature precisely where this contradiction is most binding. Forcing F_{inst} inside the translog also produces a MAPE of 53 per cent, compared with 14 per cent for C_{tec}—a 3.8× degradation. This empirically confirms the methodological decision made in Section 4.5.

5.5 H4 — Heterogeneity L9/L10 vs. L1–L5

Figure 7 illustrates the gap in average cost between the conventional lines (L1–L5) and the automated lines (L9/L10). Under C_{tec} alone the difference is bounded: the technology of the two groups is comparable in average cost per passenger, with automation shifting the labour share down without altering the order of magnitude of total operating cost per passenger. Under C_{total} the gap widens dramatically because F_{inst}, concentrated on L9/L10, adds several euros per passenger on the institutional side. The implication is that any operational efficiency gain from automation is swamped by the concession fee in the consolidated accounts—an empirical fact that has nothing to do with the technology of driverless operation as such.

Figure 7 (right) reports the operating cost shares by year, which remain stable around a labour-dominated structure (57–58 per cent), with energy at 4.6–5.9 per cent.



(a) L9/L10 vs. L1–L5: the gap is institutional.

(b) Operating cost shares (C_{tec}) by year.

Figure 7: Heterogeneity and operating cost shares. Left: stacked bars decompose the passenger-weighted average cost into C_{tec} (technological) and F_{inst} (institutional). Right: stable labour-dominated cost structure of C_{tec}.

6 Robustness

6.1 Accounting validation

The ratio of predicted to observed cost by line-year has a mean absolute percentage error (MAPE) of 14.1 per cent under C_{tec}. As discussed in Section 5.4, forcing F_{inst} inside the translog raises the MAPE to 52.9 per cent. The difference is the signature of the additive decomposition, not noise.

6.2 Sensitivity to priors

Figure 8 reports the sensitivity of RTD to the priors. The Allen–Uzawa elasticities σ and the time trend λ do not materially affect RTD (the γ_{ij} terms enter only at second order and cancel at the sample mean). The β_1 prior does affect RTD by construction: it is the direct anchor. Varying β_1 by ±0.10 shifts RTD between 1.79 and 2.79.

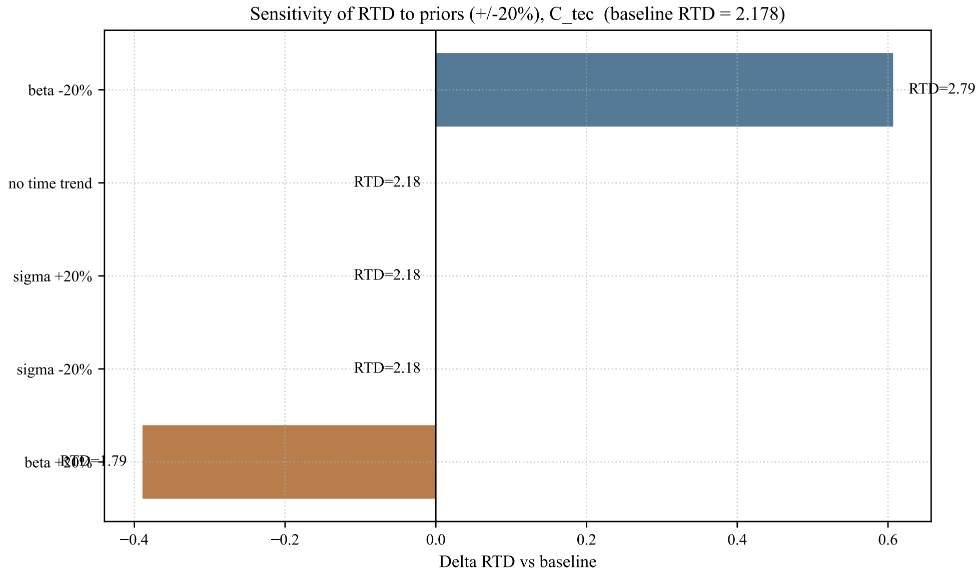


Figure 8: Sensitivity of RTD to priors (tornado plot), C_{tec}.

6.3 Alternative imputation by passengers

Redistributing all costs proportionally to passengers and re-estimating β by ordinary least squares on the re-imputed panel yields $\beta_1 = 1.000$ and $\beta_2 = 0.000$ exactly: RTD $= 1$. This is mechanical—imputation defines the elasticity—and demonstrates that the use of car-km as the imputation key is a *condition for identification*. The native OLS estimate by car-km imputation yields RTD $= -42.6$, confirming that the free regression is uninformative with this variability. Anchoring β on the literature is therefore indispensable.

6.4 Aggregate $T = 3$ model

Abandoning the line-year panel and attempting to solve $T = 3$ by finite differences year on year—two equations in two unknowns β_1, β_2 with $\lambda = 0$—yields a variability matrix X with condition number **30**.**5** and parameter estimates of $\beta_1 = +0.008$ and RTD $= +129.9$. The aggregate model is structurally ill-identified.

6.5 Monoproduct specification

Omitting Y_2 and calibrating with $\beta_1 = 0.70$ (the sum of the biproduct β's) yields a passenger-weighted RTD of 1.38, against 2.18 under the biproduct C_{tec}. The biproduct specification is therefore essential: the supplied network and realised demand are economically distinct outputs, and the monoproduct version under-estimates density economies.

6.6 Excluding 2025

Recalibrating using only 2023–2024 ($N = 14$), the input cost shares change by $+0.19$ pp (s_L), and the RTD remains within $\pm 10^{-3}$. Inclusion of the annualised 2025 data does not distort the calibration.

Taken together, the robustness analyses confirm that the central findings (RTD ≈ 2.18, MC/AC ≈ 0.46, and F_{inst} at some 29 per cent of total declared cost with zero contribution to marginal cost) do not depend on the imputation key, on the partial 2025 observations, or on the specific priors. They do depend on the additive separation between C_{tec} and F_{inst}, defended theoretically in Sections 4.5–4.6 and empirically in Section 5.4.

7 Discussion and policy implications

7.1 Closing argument on H3

Three results, taken together, frame the central regulatory message of this paper. *(i)* The institutional component F_{inst} represents around 29 per cent of FMB's total declared cost, larger than the entire energy bill and second only to operating labour. *(ii)* The concavity failure of the translog specification when F_{inst} is forced inside it formally demonstrates that the concession fee is not technological cost in any sense compatible with the dual structure of cost minimisation. *(iii)* Any assessment of FMB's operational efficiency, or any social cost–benefit analysis of L9/L10, that does not separate C_{tec} from F_{inst} is biased by construction.

7.2 Anchoring in the European regulatory framework

Directive 2012/34/EU (Directive 2012/34/EU, 2012) mandates accounting separation between infrastructure managers and operators, on the rationale that mixing infrastructure-access charges with operating costs distorts efficiency signals and biases regulatory comparisons. Urban metro systems fall outside its formal scope, but the logic applies with greater force to FMB, where the Ifercat fee—nearly a quarter of total declared cost, larger than the entire energy bill—is paid to a separate public entity that owns the infrastructure yet appears as a current operating expense in consolidated accounts. The microeconomic demonstration of this paper—that F_{inst} cannot be represented within a cost function with standard properties—provides the formal foundation for what the directive prescribes on regulatory grounds: F_{inst} must be reported separately for efficiency assessments to be meaningful, regardless of whether formal compliance is required.

7.3 Policy recommendations

Accounting separation aligned with Directive 2012/34/EU. Presenting the Ifercat fee as a current operating expense alongside labour and energy is not innocuous: it mixes a contractual infrastructure transfer with technological inputs, biasing operating-efficiency indicators and any benchmarking against peer metros. The minimal recommendation is to report the fee separately, so that downstream users can recover the technological cost function and assess efficiency on a clean basis.

Reform of the fee: a capacity charge rather than a fixed multi-year fee. The current fee is largely independent of effective use of L9/L10 capacity. Replacing it, in whole or in part, with a charge indexed to car-km or train-paths delivered would align the operator's perceived cost with actual infrastructure utilisation, generating efficiency signals consistent with network regulation (Joskow, 2008; Gómez-Lobo and Briones, 2014); a hybrid fixed-plus-variable scheme is a natural intermediate. Relatedly, the underlying infrastructure has an expected useful life above 75 years, yet the schedule front-loads cost recovery, pressuring FMB's annual P&L; reprofiling it closer to the asset life would reduce the F_{inst} share in current cost without altering the consolidated public-debt position.

Train leasing and lessons for other cities. Leasing delivers dynamic efficiency (quicker fleet renewal, risk-sharing) at the static cost of a rigid recurrent expense; greater transparency on the split between capital recovery and service would inform the regulatory debate. More broadly, any urban system financing infrastructure through off-budget debt repaid via long-term availability fees inherits the same difficulty as L9/L10: the Ifercat architecture is efficient on the financing side but consequential for the interpretation of operating cost.

7.4 Qualitative comparison with European peers

Madrid Metro operates under a different institutional architecture: the infrastructure is fully publicly owned and there is no analogous infrastructure-access fee in the operator's accounts. RATP Paris integrates infrastructure ownership and operation under a single public entity, so that infrastructure recovery is internal to the same balance sheet. Transport for London (TfL) operates under concession contracts with explicit accounting separation between infrastructure managers and operators. Among major European systems, only Barcelona's L9/L10 combines this Ifercat-style concession fee with consolidated operator accounts that present the fee as a current operating expense—hence the regulatory peculiarity of the Barcelona case.

8 Limitations

- **Sample size.** $N = 21$ is insufficient for classical econometric estimation. The anchored calibration is a systematic response, not a methodological excuse.
- **Cost imputation.** The line-level allocation is derived, not accounting. The sensitivity to the imputation key (Section 6) bounds the issue, but does not eliminate it as a source of error.
- **Calibration is not causal identification.** The priors are informed choices, not estimates. Material changes in β_1 shift RTD proportionally.
- **Over-parametrised translog.** A free translog would have 17 parameters for 21 observations; calibration brings the effective degrees of freedom down to 4–5.
- **Annualised 2025.** The $\times 12/9$ rescaling may incorporate seasonality. The robustness check on excluding 2025 shows that this is immaterial for the cost shares.
- **Heterogeneity not observed.** Farsi et al. (2006) shows that ignoring line fixed effects can bias the elasticities. Our calibration absorbs part of this through line-specific α_O adjustments, but a structured fixed-effects analysis remains pending.
- **Conservative treatment of train leasing.** The treatment of train leasing as purely institutional is conservative. If it were modelled as capital with linear depreciation, AC would change marginally but MC would remain that of C_{tec}, because capital cost is fixed in the short run.
- **No interaction in the additive decomposition.** The decomposition assumes no interaction between F_{inst} and C_{tec}. This is reasonable since the concession fee is independent of operational decisions, but a finer analysis with internal TMB accounting data could detect second-order interactions.
- **Variable-cost alternatives.** Variable-cost models with quasi-fixed inputs (Caves et al., 1981) could in principle apply, but require long time series for partial-adjustment identification. With $T = 3$, the additive decomposition is the only viable identification strategy that preserves microeconomic properties.
- **Service quality.** Frequency, waiting time, occupancy and reliability are product dimensions the operator can adjust with inputs; our model implicitly assumes quality moves proportionally with car-km (Y_2), so the reported returns to density should be read as conditional on approximately constant service quality. As argued in Section 4.6, this does not affect the central conclusion on F_{inst}: the concession fee is impermeable to all three genuine substitution margins, so its exclusion from C_{tec} is required by the technology, not by the constant-quality assumption.
- **Temporal pattern of F_{inst}.** The F_{inst} share falls from 30.2 (2023) and 30.7 (2024) to 26.8 per cent in the 2025 annualised figures. We cannot distinguish genuine reductions in the fee/leasing from faster growth in C_{tec} or artefacts of the $\times 12/9$ annualisation; the conclusions on density and the additive decomposition are unaffected, holding uniformly across years.

9 Conclusions

We have calibrated a multiproduct translog cost function for FMB-TMB on a line-year panel of 21 observations. The function satisfies the regularity tests when restricted to its technological component C_{tec}, and it leaves a formal trace of the distortion induced by the Ifercat L9 concession fee when the latter is forced inside the translog. Returns to density are substantial (RTD ≈ 2.18), marginal cost is roughly half of average cost (Boiteux optimal subsidy of approximately 54 per cent), and the institutional component F_{inst} accounts for around 29 per cent of total declared cost. The methodological contribution is the formal separation, on microeconomic grounds, of the technological cost function from the quasi-fixed contractual transfers that share the FMB consolidated accounts with it. The policy contribution is to align the empirical treatment of the Ifercat concession fee with the accounting-separation principle of Directive 2012/34/EU for the broader rail sector.

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A Derivation of RTD and marginal cost

From the translog cost function (1), the cost–passenger elasticity at a given observation (with centred variables w_L, w_E, y_1, y_2) is

$$\varepsilon_{Y_1} \;=\; \frac{\partial \ln C}{\partial \ln Y_1} \;=\; \beta_1 + \delta_{11}\, y_1 + \delta_{12}\, y_2 + \rho_{LY_1}\, w_L + \rho_{EY_1}\, w_E. \tag{5}$$

The marginal cost per passenger is $\mathrm{MC} = \partial C/\partial Y_1 = \varepsilon_{Y_1}\, C/Y_1$, average cost is $\mathrm{AC} = C/Y_1$, and hence $\mathrm{MC/AC} = \varepsilon_{Y_1}$. RTD is the reciprocal: $\mathrm{RTD} = 1/\varepsilon_{Y_1}$. At the sample mean (all centred variables zero), $\varepsilon_{Y_1} = \beta_1$.

The predicted input cost share (Shephard) is

$$s_i \;=\; \alpha_i + \sum_j \gamma_{ij}\, w_j + \sum_k \rho_{ik}\, y_k. \tag{6}$$

Concavity in prices is equivalent to negative semi-definiteness of the substitution matrix $M_{ij} = \gamma_{ij} + s_i s_j - \delta^K_{ij}\, s_i$, where δ^K_{ij} is the Kronecker delta.

B Detailed imputation keys

B.1 Car-km share (*share_ckm*)

$\text{share\_ckm}_{l,t} = \dfrac{\text{peak\_trains}_{l,t} \times \text{km\_line}_l}{\sum_{l'} \text{peak\_trains}_{l',t} \times \text{km\_line}_{l'}}$, with the denominator aggregated over all lines (including L11 and the Funicular). In 2024:

Line	peak trains	km of line	trains·km	share
L1	34	22.4	761.6	25.55%
L2	20	13.1	262.0	8.79%
L3	26	18.4	478.4	16.05%
L4	20	17.3	346.0	11.61%
L5	37	18.9	699.3	23.46%
L9/L10 North	10	18.7	187.0	6.27%
L9/L10 South	14	17.2	240.8	8.08%
Included			2,975.1	**99.81%**
L11, Funicular			5.6	0.19%

B.2 Network-km share (*share_km*)

$\text{share\_km}_l = \text{km\_line}_l / \sum_{l'} \text{km\_line}_{l'}$, with denominator 128.7 km. The key is constant over time. It is used exclusively to impute net depreciation.

B.3 Allocation of the Ifercat L9 concession fee

The fee is allocated entirely to the L9/L10 blocks. The split between North and South follows the passenger share of each block within the L9/L10 group in each year (40 per cent North / 60 per cent South in 2024).

B.4 Documented limitation of leasing imputation

Public sources do not disaggregate rolling-stock allocation by line. The imputation by car-km is the best available approximation: it assumes that leasing scales with the supply of service, which is reasonable if the fleet is reasonably balanced across types.

C Cobb–Douglas robustness benchmark

As an additional robustness benchmark, we calibrate a Cobb–Douglas restriction of the translog on C_{tec}:

$$\ln C_{\text{tec}} = \alpha_0 + \alpha_L \ln W_L + \alpha_E \ln W_E + \alpha_O \ln W_O + \beta_1 \ln Y_1 + \beta_2 \ln Y_2 + \lambda t, \tag{7}$$

with homogeneity ($\alpha_L + \alpha_E + \alpha_O = 1$) imposed and the α_i anchored on the observed cost shares, the β_j on the literature averages used in the translog calibration. Under this restriction, returns to density and the MC/AC ratio are constant across observations: RTD $= 1/\beta_1$ and MC/AC $= \beta_1$.

Table 5 reports the key metrics under both forms.

Table 5: Cobb–Douglas restriction vs. translog, C_{tec}.

Metric	Translog C_{tec}	Cobb–Douglas
RTD (passenger-weighted)	2.18	2.22
RTS (passenger-weighted)	1.43	1.43
MC/AC system	0.46	0.45
AC system (€/pax)	0.91	0.91
MC system (€/pax)	0.42	0.41
Boiteux optimal subsidy (%)	54.5	55.0

The Cobb–Douglas restriction imposes unitary input substitution, which the rail-cost literature rejects (Wheat and Smith, 2015). It is included only as a robustness benchmark. The qualitative conclusions of the paper —strong density economies (RTD well above unity) and marginal cost roughly half of average cost—are preserved under the restricted form.